Local pairs and ambient witnesses

Human Compression project: L044 continuation

Retain the actual odd-degree head setting: $G \leq \mathrm{Alt}(\Omega)$ is transitive with connected string C-marking $\rho_0, \ldots, \rho_{r-1}$, all maximal omissions intransitive, $r \geq 6$ and $n = 2r + 1$. Its first three marks are an actual transitive head on $D = \{1, \ldots, 6\}$, and

$$\tau = \rho_3 = (6\ b)\delta, \quad R = G_{\geq 3} = \mathrm{Alt}(\Omega \setminus \{1, \ldots, 5\}), \quad \rho_4|_D = 1, \quad H = G_{\geq 5}, \quad H|_{D \cup \{b\}} = 1.$$

The odd involution δ is supported outside $D \cup \{b\}$. Put $K = G_{>5}$. Assume every colour $j \geq 5$ has two eligible edges for its full ambient omission G_j, without assuming they lie in the same H-orbit. For an actual H-orbit O, its *local omission* of j is the restriction of $\langle \rho_i : i \geq 5,\ i \neq j \rangle$ to O.

Lemma 1 (The weaker assignment interface)**.** *It is impossible to assign each colour $j \geq 5$ to one actual H-orbit O_j so that all three of the following conditions hold:*

1. *O_j contains an ambient-eligible j-edge;*
2. *O_j contains two j-edges eligible for its full local omission;*
3. *K is intransitive on O_j.*

The two edges in the second condition need not be ambient eligible.

Proof. We verify the input weakening in the complete acquired odd-degree orbit-budget proofs. Their stronger assignment hypotheses are not invoked as theorem statements here. Instead we replay their proofs using the following precise interfaces.

For every nontrivial H-orbit O, let F_O be the assigned colours and $d_O = |F_O| - |O|/2$. The selected-colour bound applies to the transitive action $H|_O$, using its full local eligible pairs from condition 2. It gives $d_O \leq 0$ and makes G_{F_O} transitive on O in equality. If $5 \notin F_O$, then $G_{F_O} \leq K$ is intransitive by condition 3, so equality is impossible. This recovers all the normalized deficit inequalities, including their integral and half-integral forms. Empty assignments retain their exact deficit. The fixed-anchor proof also survives verbatim: it uses this local equality and commuting marked components, and does not need an additional ambient edge.

Every internal pair forest chooses exactly one ambient-eligible edge for each assigned colour. Condition 1 supplies it. The pair switches are provided by $\tau \in G_j$. Thus the complete internal-pair theorem, its equality path and faithful rows, and the odd-degree row exit retain their stated hypotheses. For two pairs, the same direct argument shows that an eligible edge exchanges the pair vertices; a second tail mark exchanging them would connect all four points inside G_j. Consequently the two-row proof likewise needs only condition 1 and actual K-intransitivity.

When δ exchanges H-orbits, transport the two *local* edges by the actual τ. It centralizes H and each local omission, so maps their orbit partitions bijectively between the two orbits. Local eligibility is preserved. Therefore the selected-colour bound applies to the union of their assigned colours in one orbit, with exactly the combined deficit used in the acquired proofs. The ambient witness also remains ambient, since $\tau \in G_j$.

All localization and terminal-interval counts use two ambient eligible edges *globally*, which are a standing hypothesis here. In the acquired proofs the edges of the relevant colours are shown to be confined to a displayed union of orbits: on the complementary internal orbit their pair images are trivial, and every possible edge there is already joined by τ in its ambient omission.

This confinement uses the ambient witness forest, not the local pair from condition 2. Thus the complete global localization bounds remain available.

For clarity, this accounts for every assignment use in the remaining special cases. In the first internal orbit proof, pair-path equality confines the remaining ambient edges to the companion orbit; its second fixed anchor and the literal two-point residual are unchanged. In the nonfirst deficit-two proof, the local equality on V supplies transitive G_{F_V} and the pure $\mathrm{Alt}(V)$ prefix kernel. One ambient witness per colour supplies the two-component pair forest. The kernel, faithful projection, connected marking and three-point row-growth arguments are therefore unchanged. Its final four-point contradiction uses two distinct actual edges of ρ_6 on V, which condition 2 supplies. The earlier four-point-row terminal contradiction uses the same fact. No eligibility beyond existence of these two edges is used at that step.

The actual head overlap and pure-prefix kernels depend on none of the assignment pairs. The literal exterior-transposition obstruction requires two ambient edges for each later colour globally, already assumed. These observations exhaust all uses of a second assigned ambient edge in the eight-fixed-point proof, the first and nonfirst internal proofs, the fixed-anchor corollary, and the last fixed-point budgets.

We can now replay the exhaustive normalized assembly: at seven fixed points, exchanges, the two first-orbit budgets, the nonfirst deficit-two budget and the empty internal pair are excluded by exactly those arguments. The eight-point argument is unchanged as just checked. Hence $|\operatorname{Fix}(H)| \geq 9$. The last-budget proof uses the global interval bound $f + c \leq 11$, the actual anchor which is not H-fixed, the same local deficits and internal witness forests. Its sole remaining four-point row has a terminal label by local fixed-anchor equality, and its orbit growth is disjoint from ten actual fixed points. It gives the opposite inequality $|\operatorname{Fix}(H)| < 9$. This contradiction proves the weaker-interface lemma. □

Lemma 2 (Active labels at a single ambient edge)**.** *Let O be an actual H-orbit containing exactly one ambient-eligible edge of some colour $j \geq 5$. The nonidentity restrictions of the marked generators to O have connected noncommutation diagram. Consequently their active labels form a consecutive interval containing j.*

Proof. Split that diagram into its connected components. The corresponding permutation groups commute and generate the transitive action on O. Any component not containing j centralizes $\rho_j|_O$ and lies in the actual ambient omission G_j. It sends the unique eligible edge to itself: it preserves O, commutes with the edge involution and preserves each ambient omission orbit. Its endpoints lie in different such orbits, so each is fixed individually.

A factor of a transitive commuting product which fixes one point fixes every point. Indeed, remove that factor from a word carrying the fixed point to an arbitrary point; the other factors commute with it and the removed factor fixes the initial point. Every other component would therefore be trivial on O, contradicting its definition. Only the component containing j remains. Noncommutations in a string are possible only between consecutive indices, proving the interval assertion. □

Lemma 3 (The first active label)**.** *Suppose each actual H-orbit has at most one ambient-eligible edge of colour j. On any orbit containing such an edge, the active interval of the preceding lemma starts at 5 or 6.*

Proof. An orbit on which K is intransitive has a nontrivial ρ_5-restriction, and its interval starts at 5. Consider instead a K-transitive orbit O whose first active index m is greater than 5. Suppose $m \geq 7$. The whole-orbit transport supplies an actual K-orbit $T = O^p$ inside a K-intransitive H-orbit W, with $p \in G_{<5}$. It transports the ambient j-edge and the complete marked K-action. Thus ρ_6 fixes T and $L = G_{\geq 7}$ is transitive on T.

The mark ρ_5 centralizes L, so T^{ρ_5} is an actual L-orbit. If it differs from T, they are disjoint subsets of W. Since $j \geq m \geq 7$, ρ_5 transports the witness to a second ambient-eligible j-edge in W, a contradiction. Hence ρ_5 preserves T. It preserves the unique ambient j-edge in W and fixes

its endpoints individually, since it belongs to G_j. Centralization of the transitive $L|_T$ then makes it fix T pointwise. Both K and ρ_5 preserve T, so $W = T$, contrary to the chosen K-intransitivity of W. Thus $m = 6$ is the only possibility. □

Theorem 4 (A genuine local crossing after transport)**.** *Some colour $j \geq 5$ and actual H-orbit O satisfy:*

1. *its full local j-omission has exactly two orbits, joined by exactly one j-edge;*
2. *that edge is ambient eligible, and every actual H-orbit has at most one ambient-eligible j-edge;*
3. *K is intransitive on O, and the active labels on O form an interval beginning at 5.*

There is a second, distinct H-orbit with an ambient-eligible j-edge. Its active labels form an interval beginning at 5 or 6; local uniqueness and K-intransitivity on that second orbit are not asserted.

Proof. For each colour choose an H-orbit with two ambient eligible edges if one exists, and retain both. Otherwise choose any one ambient witness edge. If K is transitive on that orbit, the complete whole-orbit transport gives an element of $G_{<5}$ carrying it, as a whole K-orbit, into an H-orbit on which K is intransitive. Transport every retained edge by the same element. A colour needing transport is greater than 5: an ambient 5-edge cannot lie in a K-transitive orbit. The transporting element commutes with the colour and lies in its ambient omission, so all retained edges remain actual and ambient eligible. The preferences are therefore preserved simultaneously.

Each final assigned orbit contains at least one edge eligible for the full local omission. If all contained two, they would satisfy the preceding weaker-interface lemma, which is impossible. Choose one with exactly one. A transitive action with exactly one edge joining different full omission orbits has exactly two such orbits: the connected graph obtained by collapsing them has just that one edge. This proves statement 1.

If any H-orbit had two ambient j-edges, the preference and simultaneous transport would have supplied two in this target, hence two locally eligible edges. This is impossible, proving 2. The target was normalized to be K-intransitive. Thus ρ_5 acts nontrivially on it; otherwise K and H would have the same transitive action. The previous lemma proves 3.

There are two ambient j-edges globally, and statement 2 puts another one in a different H-orbit. The preceding two lemmas give its active interval and its first index. We do not transport this second edge and claim its image is distinct: such transported witnesses may coincide, and no such assertion is needed. □

Attribution and boundary. The complete acquired case proofs remain bound to their original hypotheses. The first lemma explicitly checks each changed use and proves the weaker interface by their complete arguments. The new transport preference and commuting-factor arguments then supply a genuine local unique crossing, rather than inferring one from a split ambient pair. The source's local crossing and two-path strategy is retained, with these orbit distinctions made explicit. Full path counting, terminal residuals and later global attachments remain outside this note. No new induction, computation, formal, original or question credit is claimed.